

THE MINIMUM DIMENSION OF A MAXIMAL FACE OF THE 6×6 COMPLETELY POSITIVE CONE

ABSTRACT. Let $\text{low}(n)$ denote the minimum dimension of a maximal proper face of the $n \times n$ completely positive cone. We prove

$$\text{low}(6) = 7.$$

Holmgren and Zhang established the upper bound by exhibiting a 7-dimensional maximal face. The contribution here is the matching global lower bound, obtained from the classification of the extreme rays of $\mathcal{COP}^6$ and a linear-independence argument for rank-one matrices generated by zeros. This resolves the $n = 6$ instance of the even-dimensional open problem stated by Kostyukova and Tchemisova.

1. PRELIMINARIES AND STATEMENT

Let $\mathcal{S}^n$ be the space of real symmetric $n \times n$ matrices, equipped with $\langle A, X \rangle = \text{tr}(AX)$. The completely positive and copositive cones are

$$\mathcal{CP}^n = \text{cone}\{xx^\top : x \in \mathbb{R}_+^n\},$$

$$\mathcal{COP}^n = (\mathcal{CP}^n)^* = \{A \in \mathcal{S}^n : x^\top Ax \geq 0 \text{ for all } x \in \mathbb{R}_+^n\}.$$

For a maximal proper face F of $\mathcal{CP}^n$, its dimension is the dimension of its linear span, and

$$\text{low}(n) = \min\{\dim F : F \text{ is a maximal proper face of } \mathcal{CP}^n\}.$$

Theorem 1. *Every maximal proper face of $\mathcal{CP}^6$ has dimension at least 7, and $\mathcal{CP}^6$ has a maximal face of dimension 7. Consequently,*

$$\text{low}(6) = 7.$$

The problem of determining the exact value of the tight lower bound on the dimension of a maximal face of $\mathcal{CP}^n$ was raised by Dickinson [2]. Holmgren and Zhang exhibited a 7-dimensional maximal face of $\mathcal{CP}^6$ from an exposed Case 19 ray of $\mathcal{COP}^6$ and obtained the formal bound $6 \leq \text{low}(6) \leq 7$ [5, Lemma 4.7 and Theorems 4.8–4.9]. Kostyukova and Tchemisova subsequently proved $n \leq \text{low}(n) \leq n + 3$ for even $n \geq 6$ [6], and left the exact value of $\text{low}(n)$ open in that range [6, Section 5]; at $n = 6$ their estimate reads $6 \leq \text{low}(6) \leq 9$ and so does not improve on the bound above. The interval that Theorem 1 closes is therefore $\{6, 7\}$, and the new assertion is the global lower bound $\text{low}(6) \geq 7$.

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For $A \in \mathcal{COP}^n \setminus \{0\}$, put

$$F_A = \mathcal{CP}^n \cap A^\perp.$$

If $X = \sum_r x_r x_r^\top \in \mathcal{CP}^n$, then $\langle A, X \rangle = \sum_r x_r^\top A x_r$, and every summand is nonnegative. Hence

$$(1) \quad F_A = \text{cone}\{xx^\top : x \in \mathbb{R}_+^n \setminus \{0\}, x^\top A x = 0\}.$$

Lemma 2 (Extreme exposers). *Let K be a proper cone in a finite-dimensional inner-product space. Every maximal proper face F of K has the form*

$$F = K \cap y^\perp$$

for some y generating an extreme ray of K^* .

Proof. Let x lie in the relative interior of F . If $z \in K^*$ satisfies $\langle z, x \rangle = 0$, then for each $w \in F$ the point $x - \varepsilon(w - x)$ lies in F for all small $\varepsilon > 0$; since z is nonnegative on F and vanishes at x , this forces $\langle z, w \rangle = 0$. Hence the conjugate face

$$F^\Delta = K^* \cap F^\perp = K^* \cap x^\perp$$

is the set on which the linear functional $\langle \cdot, x \rangle$, nonnegative on K^* because $x \in K$, vanishes; so F^Δ is an exposed face of K^* . It is nonzero, since F is a proper face and therefore x lies on the boundary of K , so a hyperplane supporting K at x supplies a nonzero element of $K^* \cap x^\perp$.

As K^* is again a proper cone, F^Δ is pointed and hence has an extreme ray $\mathbb{R}_+ y$. Since F^Δ is a face of K^* , the ray $\mathbb{R}_+ y$ is extreme in K^* . Moreover,

$$F \subseteq K \cap y^\perp \subsetneq K,$$

so maximality gives $F = K \cap y^\perp$. $\square$

A nonzero vector $u \in \mathbb{R}_+^n$ is a *zero* of $A \in \mathcal{COP}^n$ if $u^\top A u = 0$. It is *minimal* if no zero has support strictly contained in $\text{supp } u$. We normalize zeros by $\mathbf{1}^\top u = 1$.

Lemma 3 (Zero-dimension bound). *Let $u_1, \dots, u_p$ be distinct normalized zeros of $A \in \mathcal{COP}^n$ that span $\mathbb{R}^n$. Then $\dim F_A \geq n + 1$ if either*

- (1) $p \geq n + 1$, or
- (2) $p = n$ and $u_i^\top A u_j = 0$ for some $i \neq j$.

Proof. After relabeling, let $u_1, \dots, u_n$ be linearly independent and put $U = (u_1 \cdots u_n)$. The invertible linear map

$$\Phi(X) = U^{-1} X U^{-\top}$$

sends $u_i u_i^\top$ to $e_i e_i^\top$, so these n matrices are linearly independent.

If $p \geq n + 1$, write $u_{n+1} = U c$. Since the zeros are distinct and normalized, c has at least two nonzero coordinates. Thus $\Phi(u_{n+1} u_{n+1}^\top) = c c^\top$ has a nonzero off-diagonal entry and is independent of the n diagonal matrices.

If $p = n$ and $u_i^\top A u_j = 0$, then $u_i + u_j$ is a zero and

$$\Phi((u_i + u_j)(u_i + u_j)^\top) = (e_i + e_j)(e_i + e_j)^\top,$$

which again supplies an independent off-diagonal direction. Equation (1) completes the proof. $\square$

2. PROOF OF THE MAIN THEOREM

Proof of Theorem 1. Let F be a maximal proper face of $\mathcal{CP}^6$. By Lemma 2, $F = F_A$ for a matrix A generating an extreme ray of $\mathcal{COP}^6$.

A copositive matrix is called exceptional if it is not the sum of a positive-semidefinite matrix and an entrywise nonnegative matrix. Afonin, Hildebrand, and Dickinson classified all extreme rays of $\mathcal{COP}^6$ [1, Theorem 5.1]. Up to permutation and positive diagonal congruence, such a ray is nonexceptional, is obtained by padding an exceptional extreme ray of $\mathcal{COP}^5$ with a zero row and column, or belongs to one of Cases 1–19. These congruences induce automorphisms of $\mathcal{CP}^6$ and preserve $\dim F_A$.

Nonexceptional rays. They are generated by $E_{ii} = e_i e_i^\top$, by $E_{ij} = e_i e_j^\top + e_j e_i^\top$ for $i < j$, and by $aa^\top$ with a having both positive and negative coordinates [1, Section 2]. Their zero sets give

$$\dim F_{E_{ii}} = 15, \quad \dim F_{E_{ij}} = 15 + 15 - 10 = 20, \quad \dim F_{aa^\top} = 15.$$

Indeed, the first face spans the symmetric matrices supported outside row and column i . For E_{ij} , the zero set is the union of the coordinate hyperplanes $x_i = 0$ and $x_j = 0$; the corresponding 15-dimensional symmetric subspaces intersect in a 10-dimensional subspace. Finally, the zeros of $aa^\top$ form $a^\perp \cap \mathbb{R}_+^6$, which has nonempty relative interior in the 5-dimensional space $a^\perp$, and their rank-one squares span all symmetric forms on that space.

Exceptional rays. An exceptional extreme matrix is irreducible with respect to the positive-semidefinite cone. Otherwise, for some $0 \neq P \succeq 0$ and $\varepsilon > 0$, the decomposition $A = (A - \varepsilon P) + \varepsilon P$ inside $\mathcal{COP}^6$ would force A to be positive semidefinite by extremality. Hildebrand's criterion therefore implies that the minimal zeros of A span $\mathbb{R}^6$ [4, Theorem 4.5].

Suppose first that $A = B \oplus 0$ is a zero-padded exceptional extreme matrix of order 5; this is component O5 of [3, Table 1], and it is the one component we argue directly rather than read off that table.¹ Then e_6 is a minimal zero, and every other minimal zero has sixth coordinate zero: deleting a positive sixth coordinate would otherwise produce a zero with strictly smaller support. Thus every other minimal zero u satisfies $u^\top A e_6 = 0$. Since the minimal zeros span $\mathbb{R}^6$, there are at least six. If there are at least seven, Lemma 3(1) applies; if there are exactly six, Lemma 3(2) applies to e_6 and any other minimal zero. Hence $\dim F_A \geq 7$.

It remains to treat Cases 1–19. Let $u_1, \dots, u_p$ be the normalized minimal zeros and define

$$I_\alpha = \text{supp } u_\alpha, \quad J_\alpha = \{k : (A u_\alpha)_k = 0\}.$$

¹With I_α, J_α as defined below, the O5 row records $J_\alpha \setminus I_\alpha = \emptyset$ for $\alpha \leq 5$. Since the sixth row of A vanishes, $(A u_\alpha)_6 = 0$ for every α , so $6 \in J_\alpha$ always and the row understates J_α for $\alpha \leq 5$. The argument given here uses only $A e_6 = 0$ and is independent of that row.

For any zero u of a copositive matrix, $Au \geq 0$: applying copositivity to $u + te_k$ and differentiating at $t = 0^+$ gives $(Au)_k \geq 0$. Consequently,

$$(2) \quad u_\beta^\top Au_\alpha = 0 \iff I_\beta \subseteq J_\alpha.$$

Since $u_\alpha^\top Au_\alpha = 0$ and $Au_\alpha \geq 0$, every $k \in I_\alpha$ has $(Au_\alpha)_k = 0$; thus $I_\alpha \subseteq J_\alpha$.

The extended minimal-zero support data of Hildebrand and Afonin [3, Table 1] are reproduced in Table 1, in their labelling both of the components and of the minimal zeros within each component. Their table lists 22 main components: O5 is the zero-padded case treated above, and the other 21 are those below. Only I_α and $J_\alpha \setminus I_\alpha$ are recorded, since $I_\alpha \subseteq J_\alpha$ gives $J_\alpha = I_\alpha \cup (J_\alpha \setminus I_\alpha)$; a subset of $\{1, \dots, 6\}$ is written as the string of its elements, so 146 denotes $\{1, 4, 6\}$. For each component with $p = 6$ the last column names a pair $\beta \neq \alpha$ with $I_\beta \subseteq J_\alpha$, which by (2) is the hypothesis of Lemma 3(2).

Component	p	I_α	$J_\alpha \setminus I_\alpha$	certificate
1	6	12, 13, 14, 25, 36, 456	345, 246, 23, 16, 15, $\emptyset$	$I_2 \subseteq J_1$
2	6	12, 13, 14, 25, 356, 456	345, 246, 23, 16, $\emptyset, \emptyset$	$I_2 \subseteq J_1$
3	6	12, 13, 14, 256, 356, 456	345, 24, 236, $\emptyset, \emptyset, \emptyset$	$I_2 \subseteq J_1$
4	6	12, 13, 24, 345, 156, 456	346, 26, 15, $\emptyset, \emptyset, \emptyset$	$I_2 \subseteq J_1$
5	6	12, 13, 145, 246, 346, 456	35, 256, $\emptyset, \emptyset, \emptyset, \emptyset$	$I_2 \subseteq J_1$
6	6	12, 13, 245, 345, 246, 356	34, 256, $\emptyset, \emptyset, \emptyset, 1$	$I_2 \subseteq J_1$
7	6	15, 26, 123, 234, 345, 456	24, 13, 6, $\emptyset, \emptyset, \emptyset$	$I_3 \subseteq J_2$
8	6	12, 134, 135, 246, 346, 256	36, 5, 4, $\emptyset, \emptyset, \emptyset$	$I_3 \subseteq J_2$
9.1	6	12, 134, 135, 246, 346, 456	36, $\emptyset, \emptyset, 5, \emptyset, 2$	$I_6 \subseteq J_4$
9.2	6	12, 134, 135, 246, 346, 456	356, $\emptyset, 2, \emptyset, \emptyset, \emptyset$	$I_3 \subseteq J_1$
10	6	12, 134, 135, 246, 356, 456	36, $\emptyset, \emptyset, 5, \emptyset, 2$	$I_6 \subseteq J_4$
11	6	123, 124, 125, 136, 246, 346	5, 5, 34, $\emptyset, \emptyset, 5$	$I_3 \subseteq J_2$
12	6	123, 124, 125, 136, 246, 356	$\emptyset, 5, 4, \emptyset, \emptyset, 4$	$I_3 \subseteq J_2$
13.1	6	123, 234, 345, 456, 156, 126	4, 15, 26, 3, $\emptyset, \emptyset$	$I_2 \subseteq J_1$
13.2	6	123, 234, 345, 456, 156, 126	4, 1, 6, 3, 2, 5	$I_2 \subseteq J_1$
14	7	12, 13, 14, 25, 45, 36, 56	345, 246, 235, 146, 126, 15, 234	—
15	7	12, 134, 135, 146, 256, 356, 456	34, 2, $\emptyset, \emptyset, \emptyset, \emptyset, \emptyset$	—
16	7	123, 124, 125, 136, 246, 346, 356	$\emptyset, \emptyset, \emptyset, \emptyset, \emptyset, 5, 4$	—
17	7	123, 124, 125, 136, 246, 356, 456	$\emptyset, \emptyset, \emptyset, \emptyset, \emptyset, 4, 3$	—
18	8	123, 234, 345, 145, 125, 346, 146, 126	$\emptyset, \emptyset, 6, 6, 6, 5, 5, 5$	—
19	6	345, 145, 125, 123, 156, 2346	$\emptyset, 6, \emptyset, \emptyset, 4, \emptyset$	$I_5 \subseteq J_2$

TABLE 1. The classification data needed for the lower bound, from [3, Table 1].

Thus Cases 14–18 satisfy Lemma 3(1), while every other component satisfies Lemma 3(2) by (2). It remains to check that this covers every exceptional extreme matrix, and not only the generic ones. Two features of the classification, both recorded in [3, Section 2], make this immediate.

First, the extreme matrices of $\mathcal{COP}^6$ are classified in [1] “with respect to their minimal zero support sets”. The classification therefore refines the partition by minimal zero support set, so the collection $\{I_1, \dots, I_p\}$, and in particular p , is constant on each piece (two pieces may share one collection,

as with 9.1/9.2 and 13.1/13.2). In particular the values $p = 7$ for Cases 14–17 and $p = 8$ for Case 18 hold throughout those pieces, and Lemma 3(1) applies to all of their matrices without any certificate.

Second, the sets J_α can only grow away from the generic locus. The matrices of a piece are given by analytic expressions in a scaling matrix and some angles ϕ_j [1, Section 5], and whether $(Au_\alpha)_i = 0$ depends only on the ϕ_j ; “larger complementary supports correspond to additional equality relations on ϕ_j , and the corresponding component of extreme matrices has lower dimension”. Moreover “almost all matrices in each piece belong to a single component, while the rest is located on the boundary of this component”. The component tabulated in Table 1 is thus the generic one, and since $(Au_\alpha)_i$ is continuous in the parameters, an entry vanishing on the generic locus vanishes on its closure, that is, on the whole piece. Hence at every matrix of the piece each J_α contains the set tabulated for it, and the certified inclusion $I_\beta \subseteq J_\alpha$ holds a fortiori. Cases 9 and 13 are the two that decompose into two sub-cases each, listed as 9.1, 9.2 and 13.1, 13.2 in [1, Section 3.4]: these share a minimal zero support set but are parametrized separately, and each of the four is a piece with its own main component. All four appear in Table 1 with a certificate. Therefore every exceptional extreme ray also satisfies

$$\dim F_A \geq 7.$$

We conclude that $\text{low}(6) \geq 7$.

Finally, Holmgren and Zhang proved that a Case 19 matrix generates an exposed ray of $\mathcal{COP}^6$ whose conjugate maximal face has dimension 7 [5, Lemma 4.7 and Theorem 4.8]. Hence $\text{low}(6) \leq 7$, and equality follows. $\square$

Remark 4. The proof uses the full extreme-ray classification only for the lower bound and one exposed Case 19 ray for the upper bound. It makes no exposedness claim for the other exceptional rays. The exact values of $\text{low}(n)$ for even $n \geq 8$ remain open.

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